

APPROVED

by Dean of School

_____ full name
«____» _____ 20__

SYLLABUS

Academic Year **2025-2026**
Semester **Fall**

Discipline: *iOS Development*

Volume of credits: 3 ECTS

Course/Syllabus designer: Arman S. Myrzakanurov

Instructor's e-mail	myrzakanurov.a@gmail.com
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Class Schedule	
Office Hours	
Office	

1. Course Overview

This course provides a comprehensive introduction to iOS development, focusing on building foundational skills in Swift programming, understanding the iOS environment, and developing basic applications. Students will learn about the essential tools and frameworks required to create iOS apps and will engage in hands-on projects to solidify their understanding.

Prerequisites: Basic knowledge of programming (preferably in any high-level language like Python or Java)

Postrequisites: Advanced iOS Development

2. Aims and objectives

The course aims to:

- Introduce students to the Swift programming language.
- Familiarize students with Xcode and the iOS SDK.
- Teach students how to design user interfaces using UIKit and SwiftUI.

- Develop problem-solving skills through practical app development projects.
- Prepare students for more advanced iOS development courses.

3. Learning outcomes

By the end of this course, students will be able to:

- Understand the basic syntax and concepts of the Swift programming language.
- Navigate and utilize Xcode effectively.
- Create simple iOS applications with interactive UI components.
- Implement basic navigation and data handling in an app.
- Debug and test iOS applications.
- Prepare apps for deployment on the App Store (theoretical overview).

4. Textbooks and readings

Primary textbook

1. “iOS Programming: The Big Nerd Ranch Guide” by Christian Keur & Aaron Hillegass

Supplementary textbooks

1. “Swift Programming: The Big Nerd Ranch Guide” by Matthew Mathias & John

Gallagher

2. “Developing iOS 12 Applications Using Swift” by Wallace Wang
3. “Основы разработки приложений под iOS и macOS.4-е изд. дополненное и переработанно” by Усов В.А.

5. Lesson Program

Week	Classes				
	Topic	Lecture	Laboratory	Tutorial	Textbook Chapter
1	Introduction to iOS Development	Overview of iOS & Swift	Setting up Xcode	Hello World in Swift	Chapter 1
2	Swift Basics	Variables, Constants, Data Types	Basic Syntax Practice	Conditionals & Loops	Chapter 2
3	Functions & Closures	Defining Functions	Function Labs	Understanding Closures	Chapter 3
4	Object-Oriented Programming in Swift	Classes & Structs	OOP Practices	Inheritance & Polymorphism	Chapter 4
5	Introduction to Xcode	Navigating Xcode	Simple App Creation	Xcode Tools	Chapter 5
6	Introduction to UIKit	Views & View Controllers	Building a Simple UI	Handling User Input	Chapter 5
7	Basic UIKit Controls	Working with Image, Label, Button	Creating Interactive UIs	Practice with Basic Controls	Chapter 6

8	ScrollView in UIKit	ScrollView Implementation	Customizing Scroll Behavior	Creating Dynamic Scrollable Content	Chapter 7
9	TableView in UIKit	Understanding TableView	Building a Simple TableView	Populating Data in TableView	Chapter 8
10	CollectionView in UIKit	Understanding CollectionView	Building a Simple CollectionView	Populating Data in CollectionView	Chapter 9
11	Delegate Pattern in iOS & Notification Center	Implementing Delegate Pattern	Practical Uses of Delegates	Custom Delegates in UIKit Components	Chapter 10
12	Navigation in iOS	Storyboards & Segues	Navigation Controllers	Modal & Push Navigation	Chapter 11
13	Networking in iOS	URLSession & REST APIs	Fetching Data	JSON Parsing	Chapter 12
14	Data Persistence	UserDefaults & Core Data	Saving Data Locally	Working with Files	Chapter 13
15	Final Project Presentations	Project Demos	Peer Reviews	Final Evaluation	Chapter 14

6. Course Requirements and Grades

COURSE ASSESSMENT PARAMETERS

Attendance	10 %
Activity on lessons	25 %
Midterm/endterm	25 %
Final exam	40 %
Total	100 %

No	Assessment Criteria	Weeks	Total
1	Attendance (must be more than 80%)		
2	Activity on seminars and homework	2-15	25
3	Quizzes and mid/end-terms	Midterm: Week 8; Endterm: Week 16	25
4	Final examination	Week 17	40
	Total		100

Grading policy:

Any instance of cheating or plagiarism in assignments, quizzes, or exams will result in an automatic grade of 0 for that specific work.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks; Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
	B	80-84%	
	B-	75-79%	
	C+	70-74%	
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.
	C-	60-64%	
	D+	55-59%	
	D	50-54%	

7. General Rules

Academic Integrity

Students are expected to adhere to the university's academic integrity policy, ensuring all work submitted is their own.

Attendance policy

Students must maintain at least 80% attendance to qualify for the final exam.